

Problems Overview

Introduction

The Port of Dover is the UK’s busiest international ferry port, linking southeast England to France across a 34 km route [1]. It handles around 9 million passengers and over 2 million freight vehicles annually [2], with ferries departing roughly every 30 minutes [3].

The port moves approximately £144 billion of goods annually, nearly one-third of UK–EU traded goods [4], and carries about 31% of all seaborne HGV traffic [4]. Its fast cross-channel operations generate an estimated £3 billion in annual transport cost savings [4] while supporting significant regional employment and economic activity in Kent [5].

Thoughts. Dover operates a high-throughput, low-buffer system in which stochastic arrivals, fixed ferry schedules, and capacity-limited border processing interact tightly. Even small fluctuations in demand or service times can push utilisation above capacity, making queue formation structurally unavoidable during peaks. Congestion is therefore not merely a result of disruption but an inherent feature of a system operating close to saturation.

Stakeholders

- **Dover Harbour Board:** Oversees port management and operations [1].
- **Ferry Operators:** P&O Ferries, DFDS, and Irish Ferries provide cross-channel services [3].
- **Border Agencies:** UK Border Force and French Police Aux Frontières conduct juxtaposed checks [6].
- **Local and Transport Authorities:** Kent Police, National Highways, and the Kent Resilience Forum coordinate traffic and resilience planning [7].

Queuing Process

Vehicles pass through check-in and passport control before entering marshalling lanes for boarding [1]. Traffic flow is monitored via real-time sensor systems [8]. During peak periods, external controls help manage congestion:

- **Dover TAP:** Temporarily holds HGVs on the A20 to prevent town-centre congestion [9].
- **Operation Brock:** Utilises a section of the M20 as a controlled queuing zone during severe disruption [7].

Bottlenecks

Delays arise when demand exceeds processing capacity, driven by weather, ferry disruption, or extended post-Brexit passport checks. Processing times range from roughly 90 seconds per car to up to 20 minutes per coach [10]. These constraints have produced severe delays, including six-hour queues for cars in 2022 and up to 18-hour queues for coaches in 2023 [11, 10], with significant spillover impacts on local roads and businesses [5].

Parameter	Meaning	Public Avail- ability	Source / Method	Impact
$\lambda(t)$	Vehicle arrival rate (time-dependent)	Partial (aggre- gated only)	Dover annual stats; DFT hourly traffic counts	Critical
c	Number of active pro- cessing booths	Yes	Port layout; operator information	High
μ	Average service rate per booth	Partial (not published)	Academic RoRo studies; inferred from processing times	Very critical
ρ	Utilisation factor $\lambda/(c\mu)$	Derived	Computed from ob- served λ and esti- mated μ	Very critical
W_q	Expected waiting time	Derived	M/M/c approxima- tion formula	High
L_q	Expected queue length	Derived	Little's Law	High
B_f	Ferry lane-metre capac- ity	Yes	DFDS / P&O vessel specifications	Moderate
α_v	Lane-metre usage per vehicle type	Yes	Standard vehicle di- mensions	Low
τ_f	Ferry departure times	Yes	Public ferry timeta- bles	Moderate

Table 1: Model parameters, public availability, and estimated impact on accuracy.

The parameters designated as “critical” or “very critical” directly govern the relationship between stochastic demand and processing capacity, and therefore exert the strongest influence on queue formation. Arrival rates $\lambda_v(t)$ determine the instantaneous load placed on the system, while service rates $\mu_{s,v}$ and inspection probabilities $p_{inspect,v}$ define the effective throughput and the likelihood that vehicles experience additional processing. Because waiting times in G/G/c queues increase superlinearly as utilisation approaches one, even small misestimations of these parameters can produce disproportionately large deviations in predicted queue lengths and delays. In contrast, geometric parameters such as lane length or vehicle density primarily constrain spillback and have comparatively limited effect on core queue dynamics.

Mathematical Formulation

To capture the formation of queues at the Port of Dover, we use a standard multi-server queueing model, approximating the time-dependent flow with an average rate λ (vehicles per hour), and each processing booth serves vehicles at an average rate μ (vehicles per hour). With c booths, the total processing capacity is therefore $c\mu$.

Utilisation

The key performance indicator for congestion is the utilisation factor:

$$\rho = \frac{\lambda}{c\mu}.$$

When $\rho < 1$, the system has spare capacity. As ρ approaches 1, the system becomes highly sensitive to fluctuations in arrivals or service times. When $\rho \geq 1$, demand temporarily exceeds capacity and queues grow rapidly.

Waiting Time Approximation

Using a standard $M/M/c$ approximation, the expected waiting time in queue is:

$$W_q \approx \frac{P(wait)}{c\mu - \lambda},$$

where $P(wait)$ is the probability that an arriving vehicle finds all booths busy. This expression illustrates that waiting times increase sharply as λ approaches $c\mu$.

Queue Length

By Little's Law, the expected queue length is:

$$L_q = \lambda W_q.$$

This mathematic model captures the essential interaction between arrival rate, processing capacity, and congestion, and provides a tractable framework for explaining delays at the Port of Dover.

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